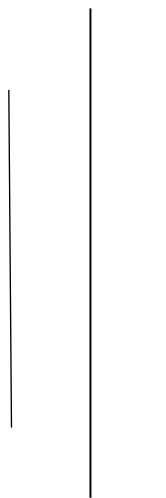




PROJECT REPORT ON
“DOCUMENT MANAGEMENT SYSTEM”



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ABSTRACT

The Document Management System is a web-based application designed to efficiently store, organize, manage, and retrieve digital documents within an organization. Traditional document handling methods often lead to issues such as document loss, duplication, security risks, and difficulties in accessing information. The proposed system addresses these challenges by providing a centralized platform for document storage and management. The system enables users to upload, view, search, download, and manage documents through a secure and user-friendly interface. Role-based access control ensures that users can only access documents according to their permissions, enhancing security and confidentiality. Advanced search and filtering capabilities allow users to quickly locate required documents, improving productivity and reducing time spent on manual document retrieval. The Document Management System aims to streamline document-related processes, improve collaboration, enhance security, and provide a reliable solution for managing organizational documents in a digital environment.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
DMS	Document Management System
JPA	Java Persistence API
JSON	JavaScript Object Notation
JWT	JSON Web Token
REST	Representational State Transfer
SQL	Structured Query Language

CHAPTER 1: INTRODUCTION

1.1. Introduction

The Document Management System (DMS) is a full-stack enterprise web application that enables organizations to securely upload, store, organize, share, version, and manage documents with complete role-based access control, audit logging, and user lifecycle management.

The system is designed for organizations that require compliance-grade document control every action is recorded in an immutable audit trail, access is enforced at the API level for every request, and user accounts go through a structured lifecycle from registration approval to eventual termination.

1.2. Objectives

The primary objectives of the Document Management System are:

- To provide a centralized repository for storing and managing organizational documents.
- To enable secure document upload, download, and sharing.
- To implement role-based access control for enhanced security.
- To maintain document version history and audit logs.
- To improve document search and retrieval efficiency.
- To reduce document redundancy and loss.
- To streamline organizational document workflows.

1.3. Scope

The Document Management System supports the following functionalities:

- User registration and authentication.
- Role-based user management.
- Document upload and storage.
- Document categorization and organization.
- Advanced search and filtering.
- Document download and sharing.
- Audit logging and activity tracking.
- User lifecycle management.

CHAPTER 2: REQUIREMENT ANALYSIS

2.1. Functional Requirements

User Management

- User Registration
- User Login
- User Profile Management
- User Approval and Deactivation

Document Management

- Upload Documents
- Download Documents
- View Documents
- Delete Documents
- Edit Document Metadata

Search and Filter

- Search by Document Name
- Search by Category
- Search by Upload Date
- Search by Keywords

Security Management

- Role-Based Access Control
- Authentication and Authorization
- Audit Trail Management

Administrative Functions

- User Management
- Role Management
- System Monitoring

2.2. Non-Functional Requirements

Performance

The system should provide fast document retrieval and response times.

Security

All sensitive operations must be protected using JWT authentication and authorization mechanisms.

Reliability

The system should ensure document availability and data consistency.

Scalability

The application should support increasing numbers of users and documents.

Maintainability

The system should be modular and easy to maintain.

Usability

The user interface should be intuitive and user-friendly.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGNS

3.1. USE CASE

The Use Case Diagram illustrates interactions between system users and the Document Management System.

Actors

- Administrator, Manager, Employee

Major Use Cases

- Login, Upload Document, Download Document, Search Document, Manage Users, Manage Roles, View Audit Logs, Manage Categories

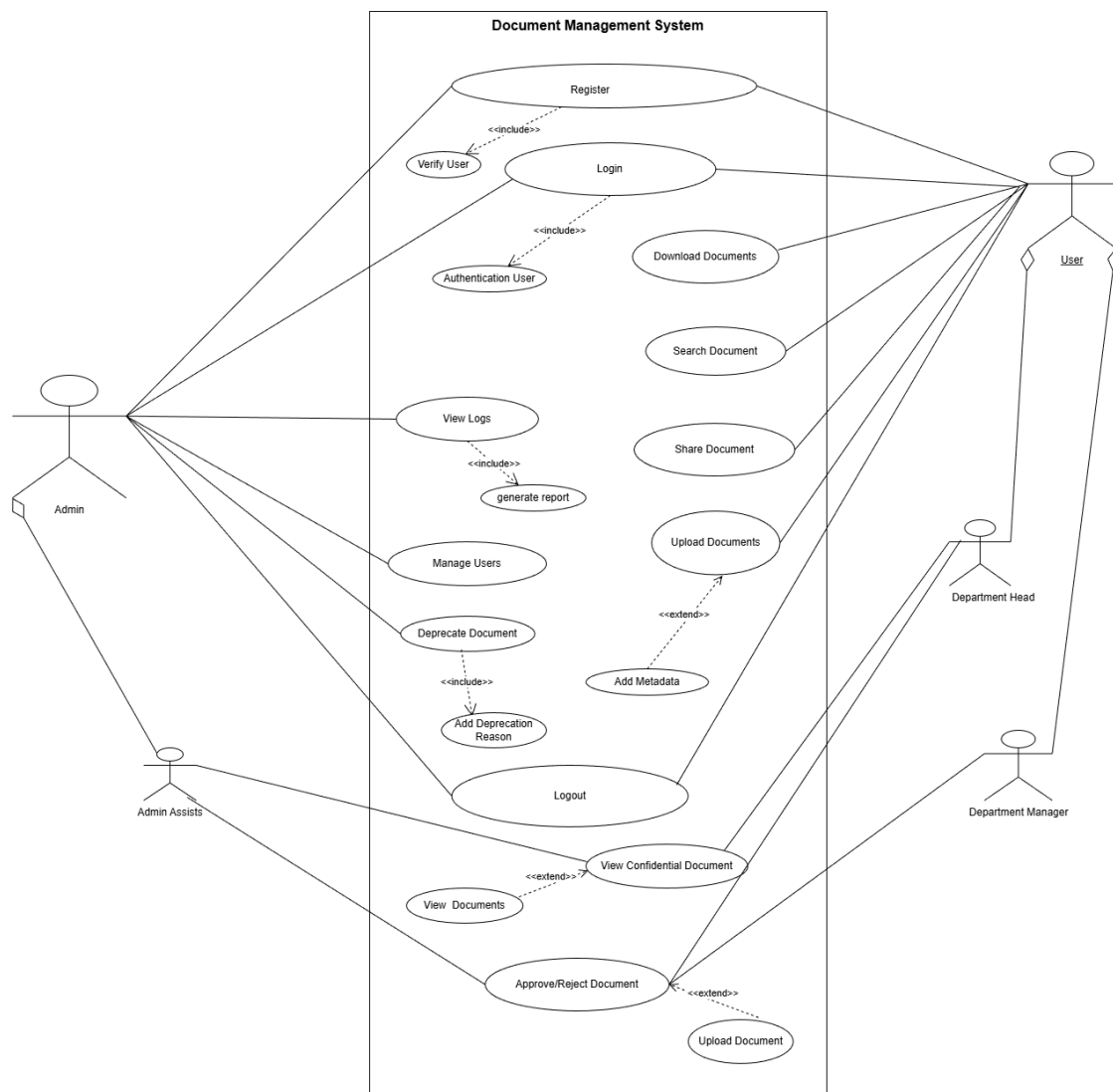


Fig. 1: Use Case Diagram

3.2. DATA FLOW DIAGRAM

Level 0 DFD: -

The Context Diagram represents the interaction between users and the Document Management System. Users submit requests such as document uploads, searches, and downloads while the system processes and returns responses.

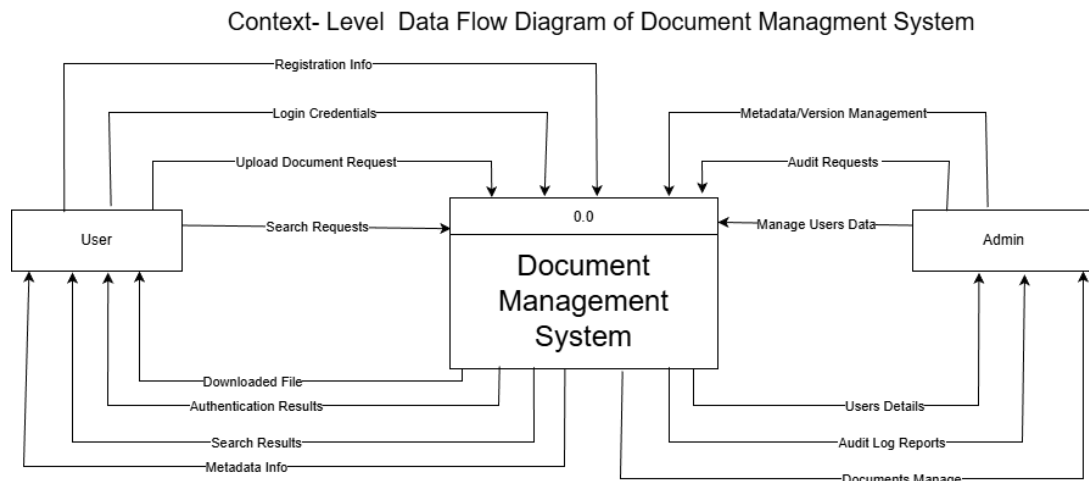


Fig. 2: Context Diagram

Level 1 DFD

The Level 1 DFD decomposes the system into major processes:

- User Authentication, Document Management, Search Management, User Administration, Audit Logging

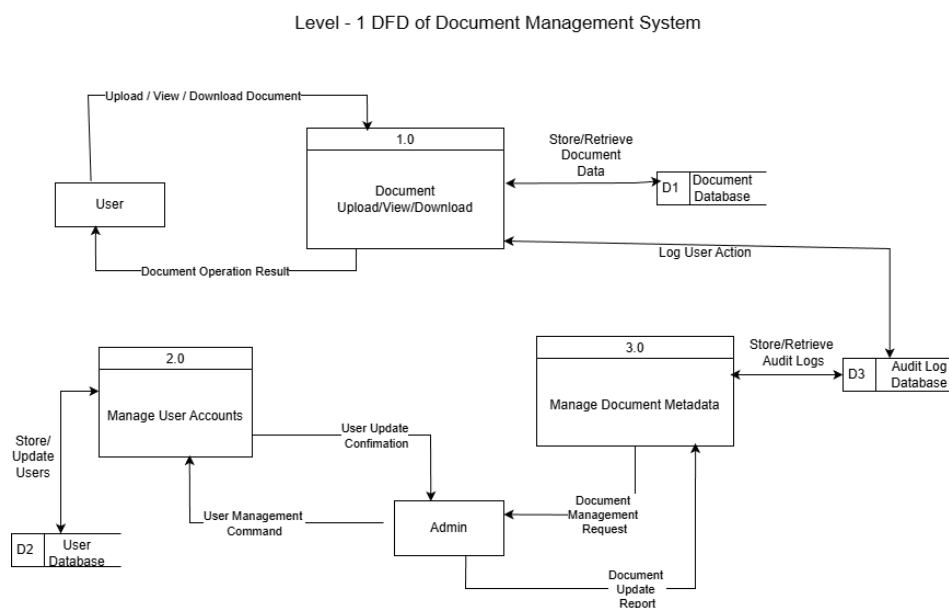


Fig. 3: Level-1 Data Flow Diagram

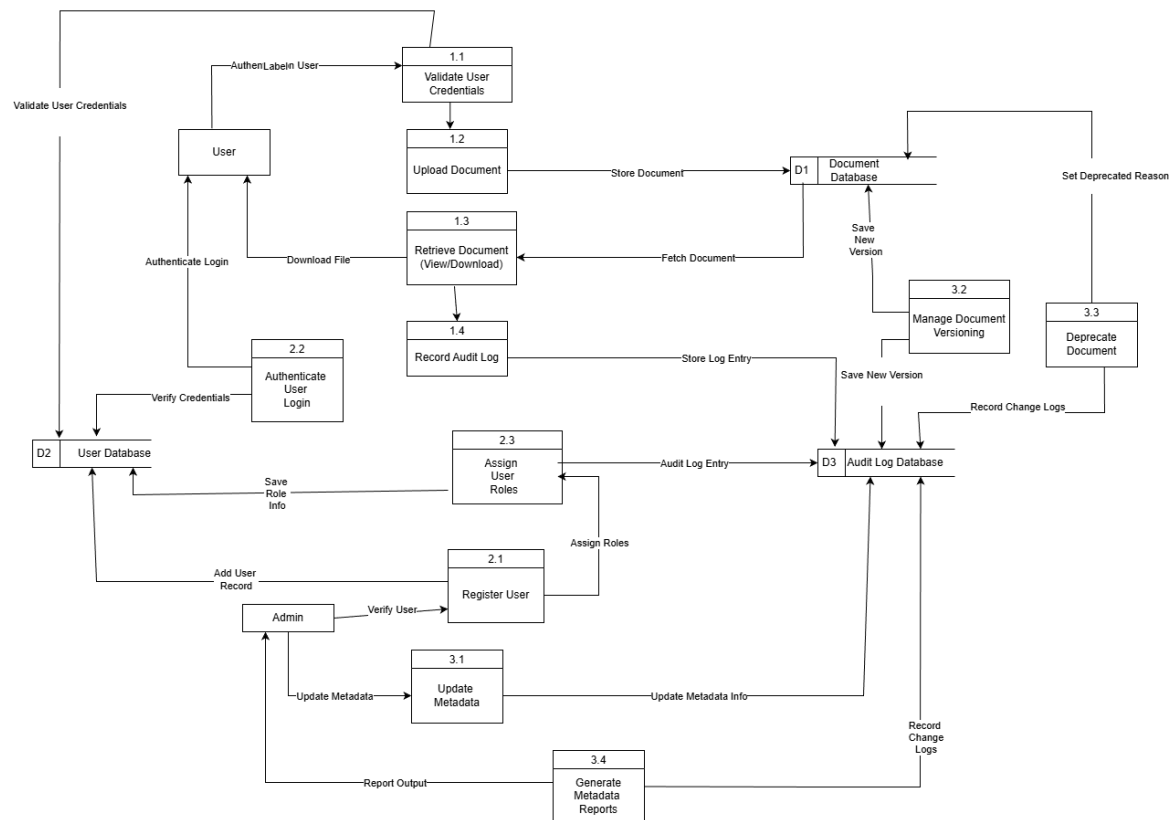


Fig. 4: Level-2 Data Flow Diagram

3.3 ENTITY RELATIONSHIP DIAGRAM

The ER Diagram illustrates relationships among system entities.

Main Entities

- Users, Roles, Documents, Metadata, Audit Logs

Relationships:

- A User belongs to one Role.
- A User can upload multiple Documents.
- A Document includes metadata.
- A User generates multiple Audit Logs.

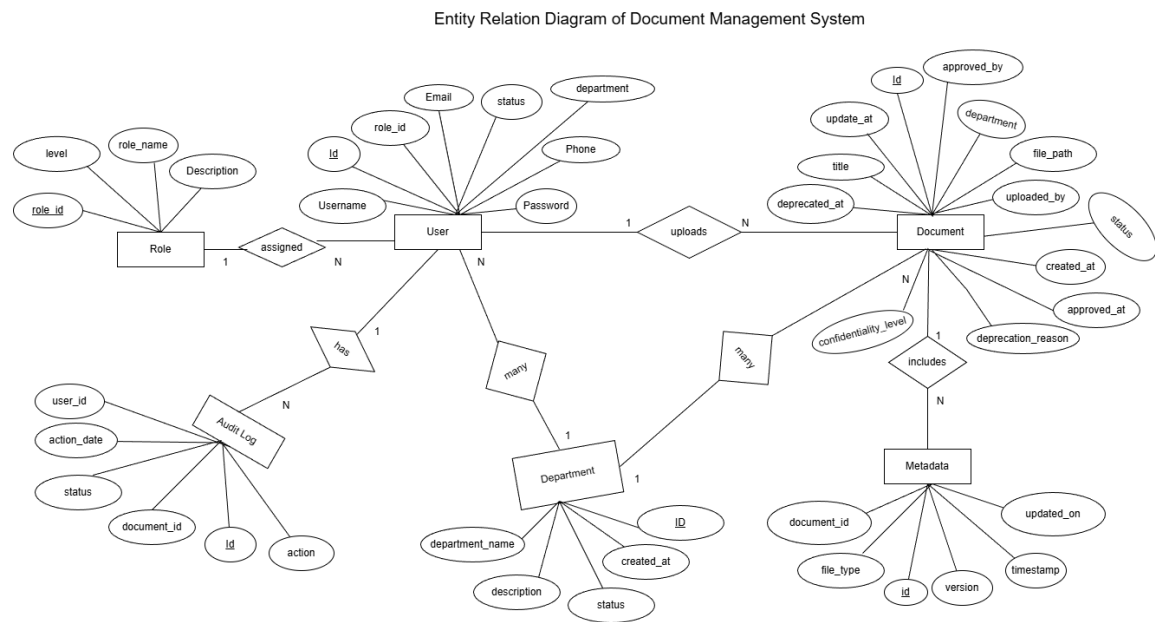


Fig. 5: Entity Flow Diagram

3.4. CLASS DIAGRAM

The Class Diagram represents the object-oriented structure of the application.

Main Classes:

- User, Role
- Document
- Category
- Audit Log
- Authentication Service
- Document Service

Class Diagram of Document Management System

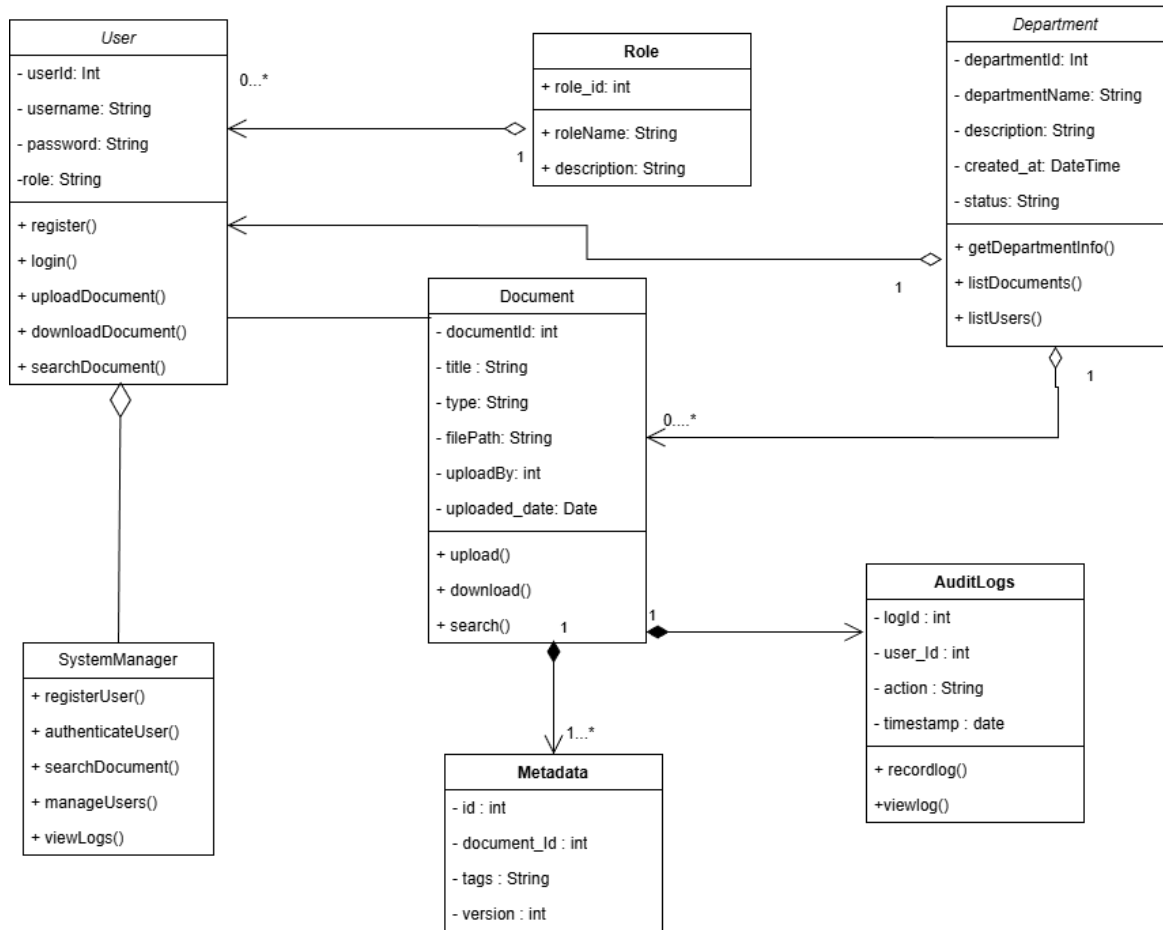


Fig. 6: Class Diagram

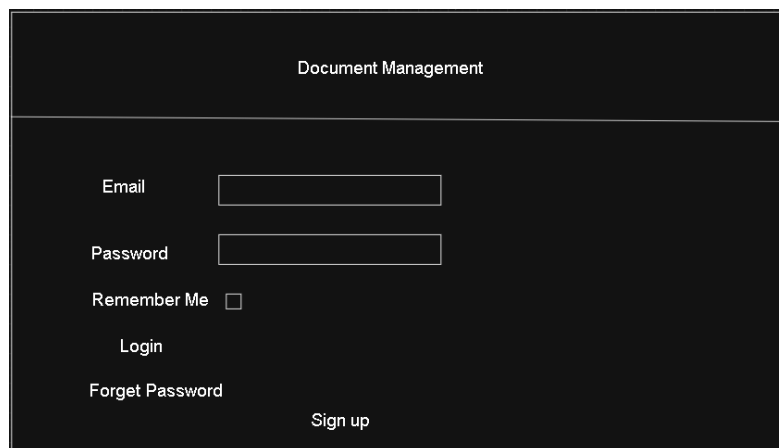
CHAPTER 4: USER INTERFACE DESIGN

4.1 Login Page

The Login Page allows authorized users to access the system using valid credentials.

Features:

- Username/Email Input, Password Input, Login Button



The login page design features a dark blue header with the text "Document Management". Below the header, there is a light blue background area containing the login form. The form includes an "Email" input field, a "Password" input field, a "Remember Me" checkbox, a "Login" button, a "Forget Password" link, and a "Sign up" link.

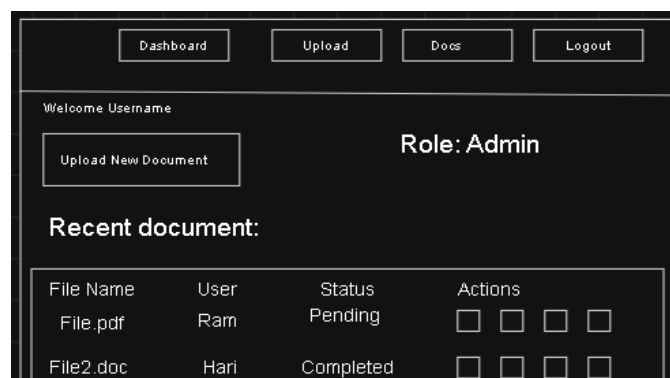
Fig. 7: Login page design

4.2 Dashboard page

The Dashboard provides an overview of system activities and document statistics.

Features:

- Document Count, User Statistics, Recent Activities



The dashboard page design features a dark blue header with four buttons: "Dashboard", "Upload", "Docs", and "Logout". Below the header, there is a light blue background area. The dashboard includes a "Welcome Username" message, a "Role: Admin" label, an "Upload New Document" button, and a "Recent document:" section. The "Recent document:" section contains a table with the following data:

File Name	User	Status	Actions
File .pdf	Ram	Pending	☐ ☐ ☐ ☐
File2.doc	Hari	Completed	☐ ☐ ☐ ☐

Fig. 8: Dashboard page design

4.3 Document Upload Page

This page enables users to upload and categorize documents.

Features:

- File Selection, Category Selection, Description Entry

My Document				
Name	Type	Date	Actions	Fliter
report.pdf	PDF		☐ ☐ ☐	
Notes.docx	DOC		☐ ☐ ☐	

Fig. 9: Document Upload page design

4.4 Document filter Page

Displays all available documents based on user permissions.

Features:

- Search & filter Functionality

Search:

Filter:

When Clicked

Fliter Document

Status

☐ ALL
 ☐ Pending
 ☐ Completed
 ☐ Rejected

Year:

2026 

Apply

Reset

Fig. 10: Document Filter page design

CHAPTER 5: IMPLEMENTATION

5.1 Development Environment

Frontend: - React.js & Tailwind CSS

Backend: - Spring Boot

Database: - PostgreSQL

Authentication: - JWT Authentication

5.2 Frontend Implementation

The frontend was developed using React.js and Tailwind CSS. It communicates with backend APIs through RESTful services.

Implemented Modules:

- Authentication Module, Dashboard Module, Document Management Module, Search Module, User Management Module

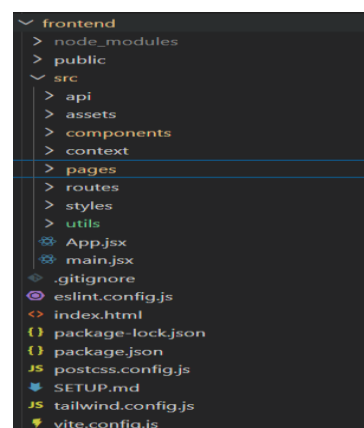


Figure 11: Frontend structure

5.3 Backend Implementation

The backend was developed using Spring Boot following a layered architecture.

Layers:- Controller Layer, Service Layer, Repository Layer, Security Layer

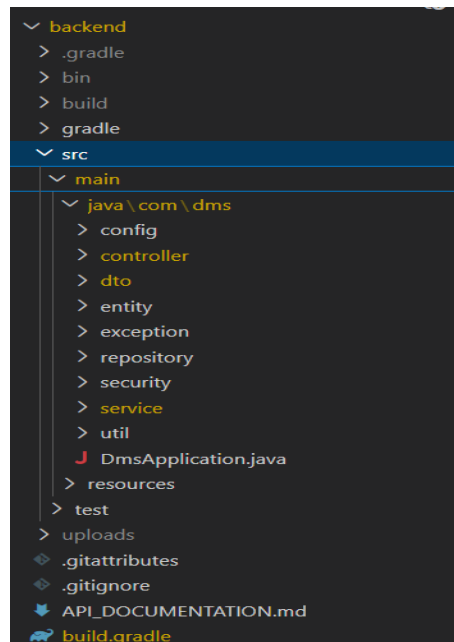


Figure 12: Backend structure

5.4 Database Implementation

PostgreSQL was used for persistent data storage.

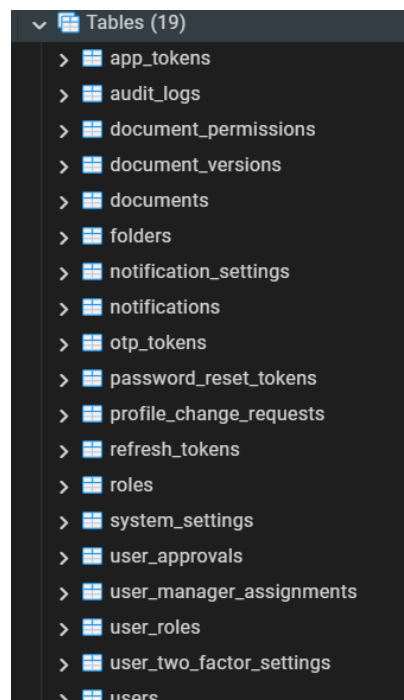


Figure 13: Database Schemas

5.5 Security Implementation

Security features include:

- JWT Authentication

- Password Encryption
- Role-Based Authorization
- Protected API Endpoints

```
public class JwtAuthenticationFilter extends OncePerRequestFilter {
    private final JwtUtil jwtUtil;
    private final CustomUserDetailsService userService;
    @Override
    protected void doFilterInternal(
        @NonNull HttpServletRequest request,
        @NonNull HttpServletResponse response,
        @NonNull FilterChain filterChain
    ) throws ServletException, IOException {
        final String authHeader = request.getHeader("Authorization");
        if (authHeader == null || !authHeader.startsWith("Bearer ")) {
            filterChain.doFilter(request, response);
            return;
        }
        try {
            final String jwt = authHeader.substring(7);
            final String userEmail = jwtUtil.extractUsername(jwt);
            if (userEmail != null &&
                SecurityContextHolder.getContext().getAuthentication() == null) {
                UserDetails userDetails =
                    userService.loadUserByUsername(userEmail);
                if (jwtUtil.isTokenValid(jwt, userDetails)) {
                    UsernamePasswordAuthenticationToken authToken =
                        new
                            UsernamePasswordAuthenticationToken(userDetails, null,
                                userDetails.getAuthorities());
                    authToken.setDetails(
                        new
                            WebAuthenticationDetailsSource().buildDetails(request));
                    SecurityContextHolder.getContext().setAuthentication(authToken);
                }
            }
        }
    }
}
```

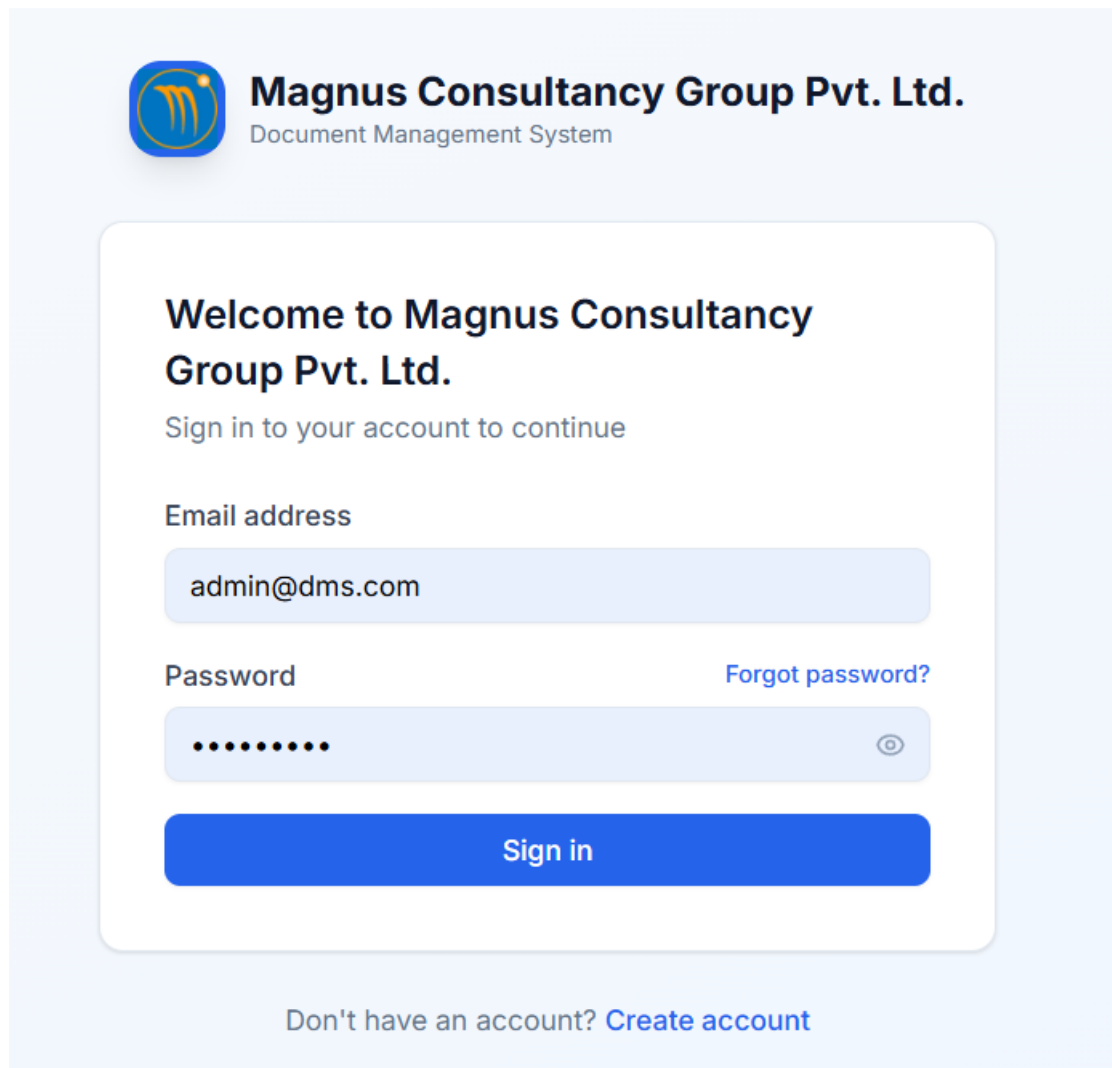
REFERENCES

- [1] C. Walls, *Spring Boot in Action*, 2nd ed. Manning Publications, 2022.
- [2] L. Richardson and M. Amundsen, *RESTful Web APIs*. O'Reilly Media, 2013.
- [3] H. Schildt, *Java: The Complete Reference*, 11th ed. McGraw-Hill Education, 2018.
- [4] M. Harrison, *PostgreSQL Fundamentals*. Packt Publishing, 2021.

APPENDICES

Screenshots

Login Page



The screenshot displays the login interface for the Magnus Consultancy Group Pvt. Ltd. Document Management System. At the top left is the company logo, a blue circle with a stylized orange 'M' and a yellow star. To its right, the company name 'Magnus Consultancy Group Pvt. Ltd.' is written in bold, with 'Document Management System' in a smaller font below it. The main content area is a white rounded rectangle with a blue border. It begins with a welcome message 'Welcome to Magnus Consultancy Group Pvt. Ltd.' followed by the instruction 'Sign in to your account to continue'. Below this are two input fields: 'Email address' containing 'admin@dms.com' and 'Password' containing a masked password of ten dots. A 'Forgot password?' link is positioned to the right of the password field. A large blue 'Sign in' button is centered below the inputs. At the bottom of the white box, a link 'Don't have an account? Create account' is provided.

Magnus Consultancy Group Pvt. Ltd.
Document Management System

Welcome to Magnus Consultancy Group Pvt. Ltd.
Sign in to your account to continue

Email address
admin@dms.com

Password
.....
[Forgot password?](#)

Sign in

Don't have an account? [Create account](#)

Sign Up Page

**Magnus Consultancy Group Pvt. Ltd.**
Document Management System

Create your account

Join Magnus Consultancy Group Pvt. Ltd. to manage your documents securely

First name

John

Last name

Doe

Username

johndoe

Email address

you@company.com

Password

Min. 10 characters

Confirm password

Repeat password

Department

Other

Create account

Already have an account? [Sign in](#)

Admin Dashboard

Magnus Consultancy Gr...

vt. 8

MAIN

Dashboard

Documents

Advanced search

TEAM

Users

Change requests

Approvals

ADMIN

Roles

Audit trail

Settings

ACCOUNT

Notifications

My profile

Dark mode

SA System Admin
admin@cms.com

Welcome back, System 🤖

System overview and admin tools

Admin

TOTAL DOCUMENTS

3

0 new this month

ACTIVE USERS

5

5 total

STORAGE USED

921.8 KB

0% of 10 GB

PENDING APPROVALS

0

Recent documents

View all →

sadhanasudhaCv

Rajal Munankarmi - 23 days ago

107 KB

multi

Rajal Munankarmi - about 1 month ago

173.2 KB

Nagarik app

Sadhana Sudha - about 2 months ago

641.6 KB

Admin quick actions

Upload

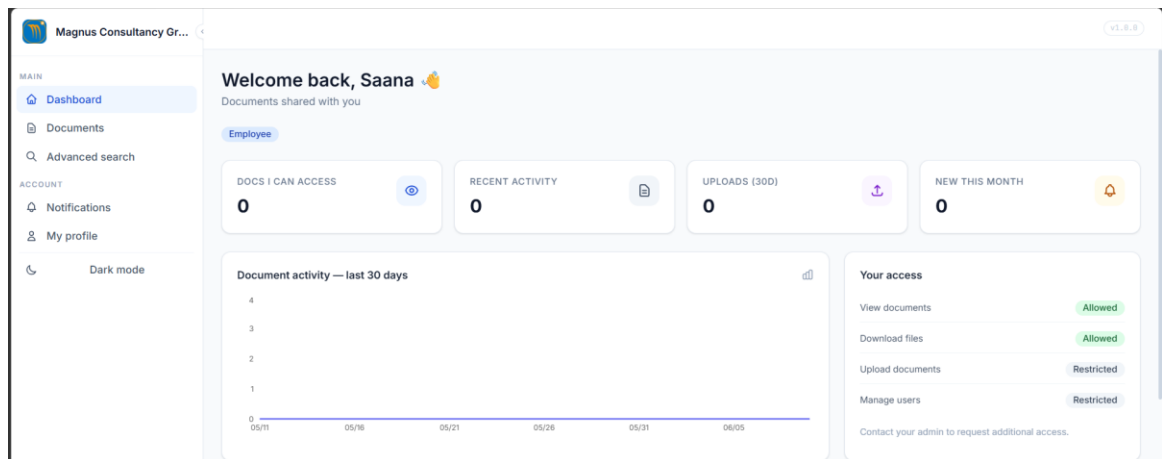
Manage users

Audit trail

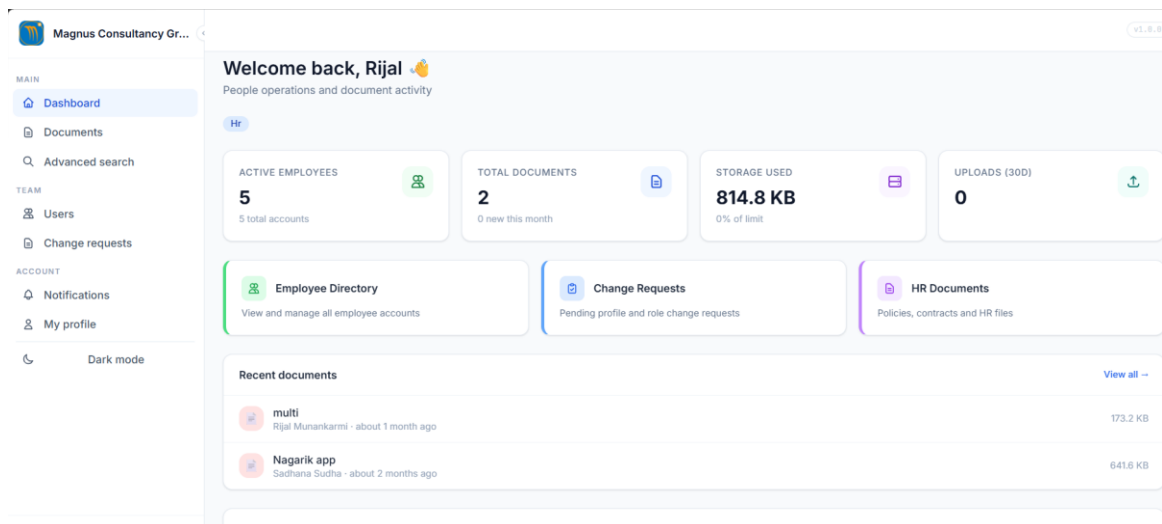
System settings

Roles & permissions

Employee Dashboard



HR Dashboard



Account Dashboard

